Thinking in Systems

A Primer

Donella H. Meadows

Edited by Diana Wright, Sustainability Institute

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— A Primer —

Audiobook Supplement

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LIST OF FIGURES

Appendix

Figure 1. How to read stock-and-flow diagrams. In this book, stocks are shown as boxes, and flows as arrow-headed "pipes" leading into or out of the stocks. The small T on each flow signifies a "faucet;" it can be turned higher or lower, on or off. The "clouds" stand for wherever the flows come from and go to—the sources and sinks that are being ignored for the purposes of the present discussion.

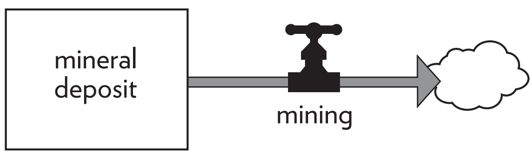


Figure 2. A stock of minerals depleted by mining.

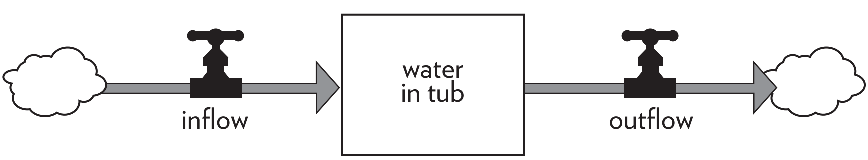


Figure 3. The structure of a bathtub system—one stock with one inflow and one outflow.

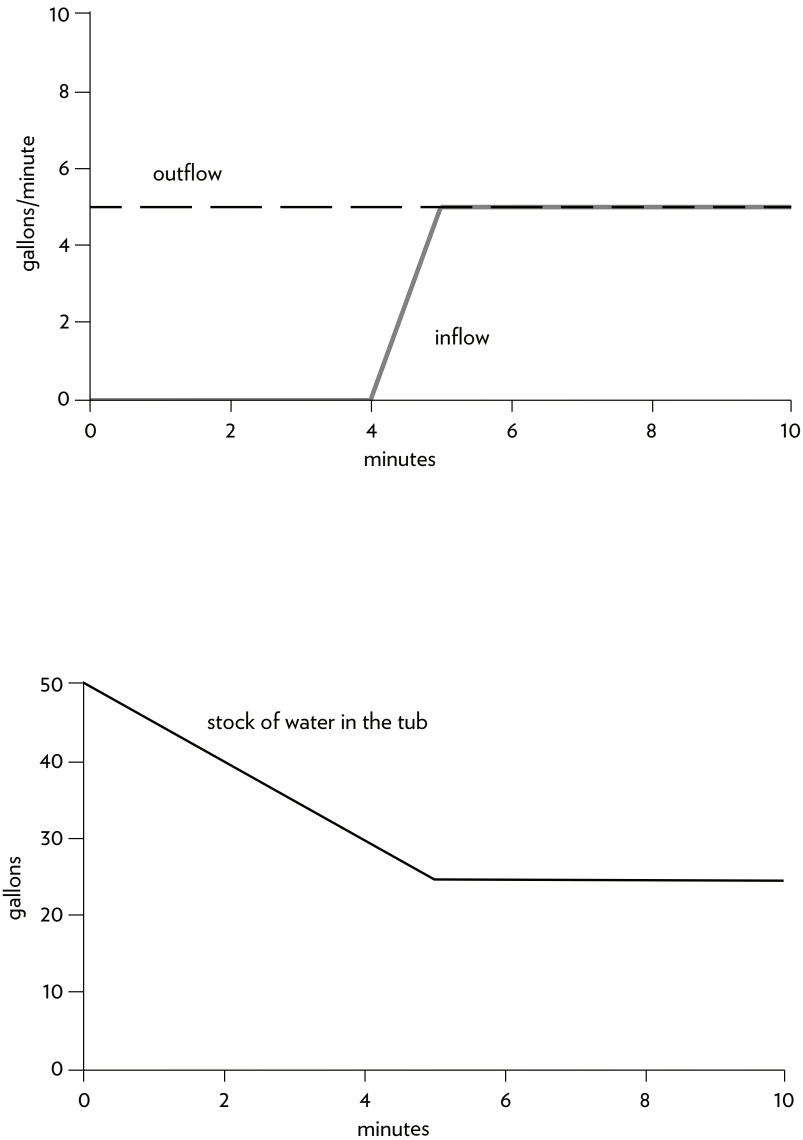


Figure 4. Constant outflow, inflow turned on after 5 minutes, and the resulting changes in the stock of water in the tub.

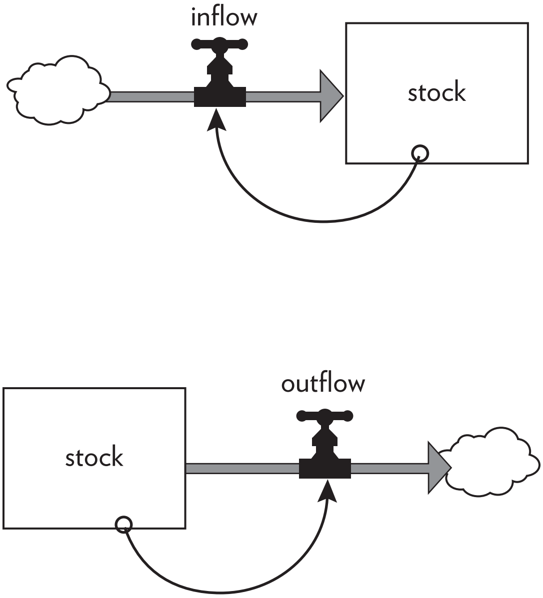


Figure 5. How to read a stock-and-flow diagram with feedback loops. Each diagram distinguishes the stock, the flow that changes the stock, and the information link (shown as a thin, curved arrow) that directs the action. It emphasizes that action or change always proceeds through adjusting flows.

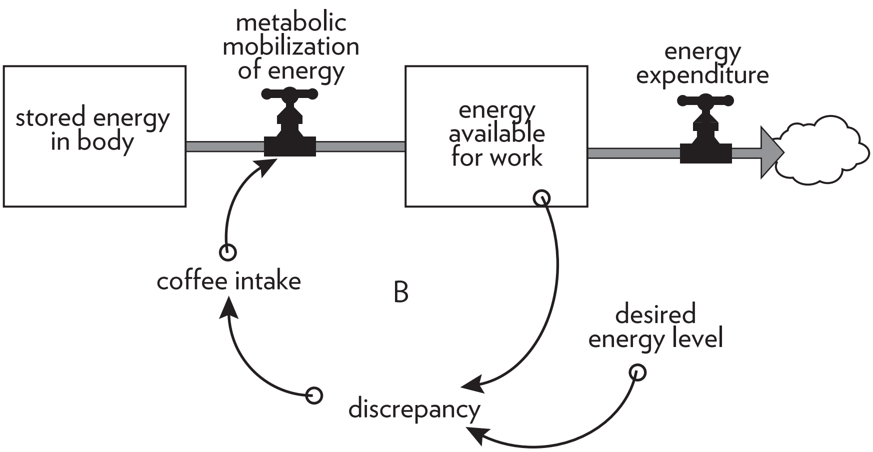


Figure 6. Energy level of a coffee drinker.

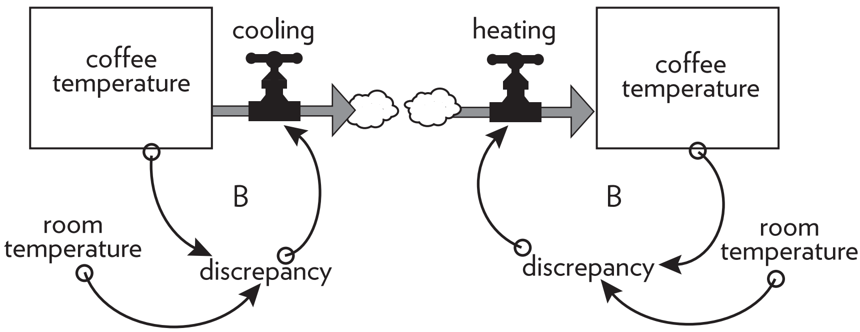


Figure 7. A cup of coffee cooling (*left*) or warming (*right*).

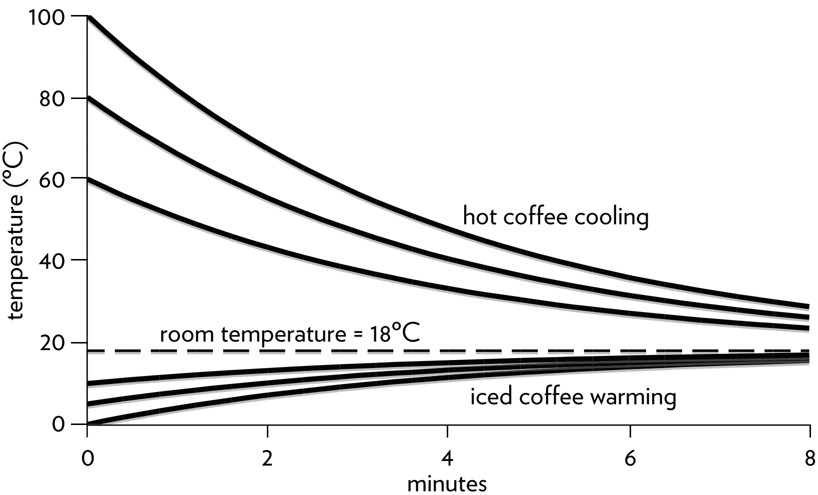


Figure 8. Coffee temperature as it approaches a room temperature of 18 °C.

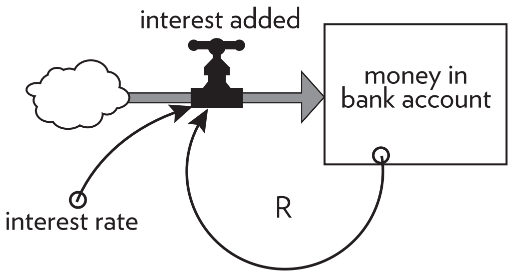


Figure 9. Interest-bearing bank account.

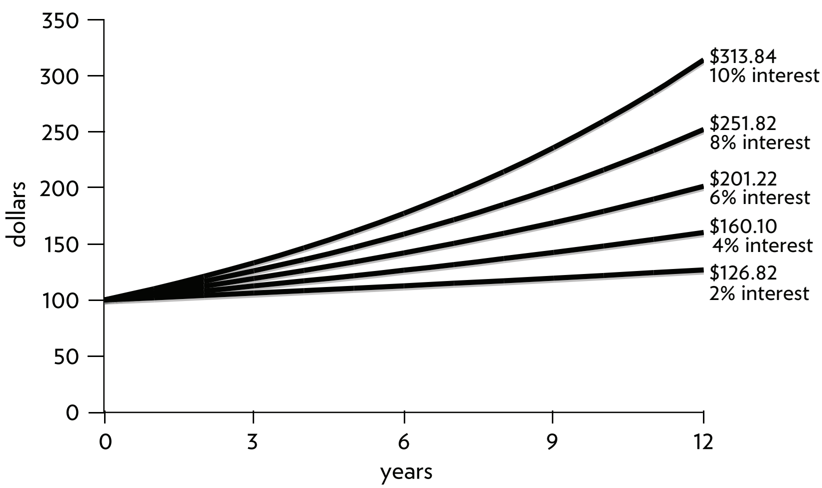


Figure 10. Growth in savings with various interest rates.

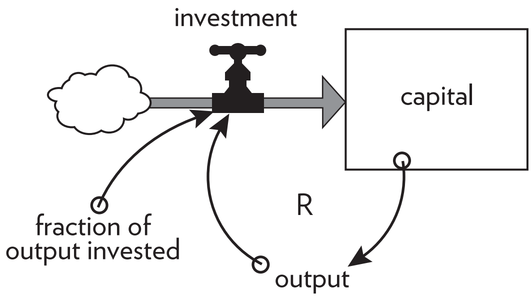


Figure 11. Reinvestment in capital.

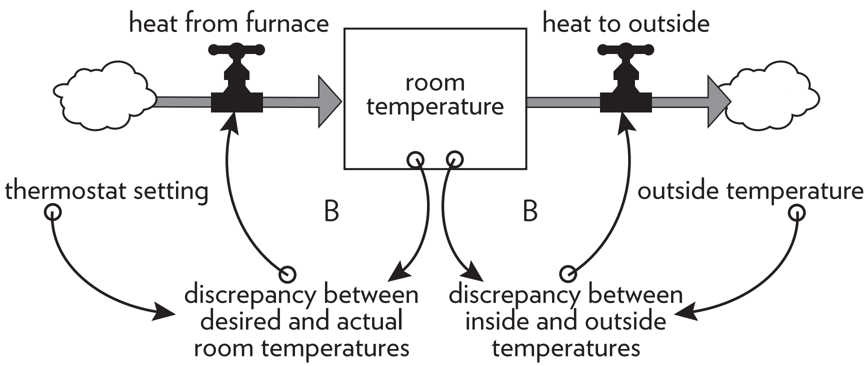


Figure 12. Room temperature regulated by a thermostat and furnace.

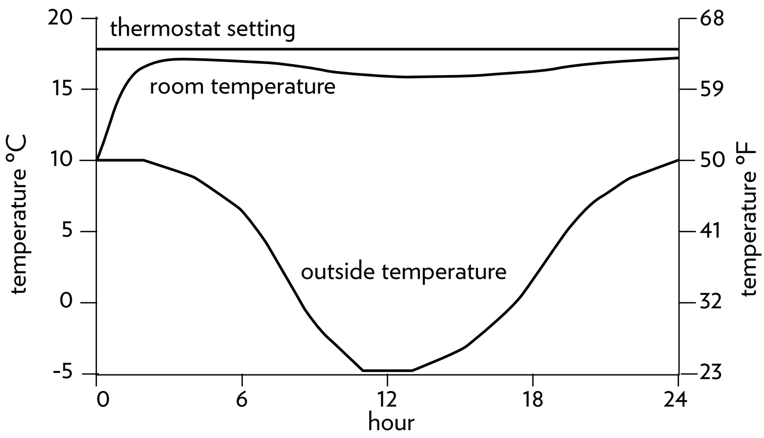


Figure 13. The furnace warms a cool room, even as heat leaks from the room and outside temperatures drop below freezing.

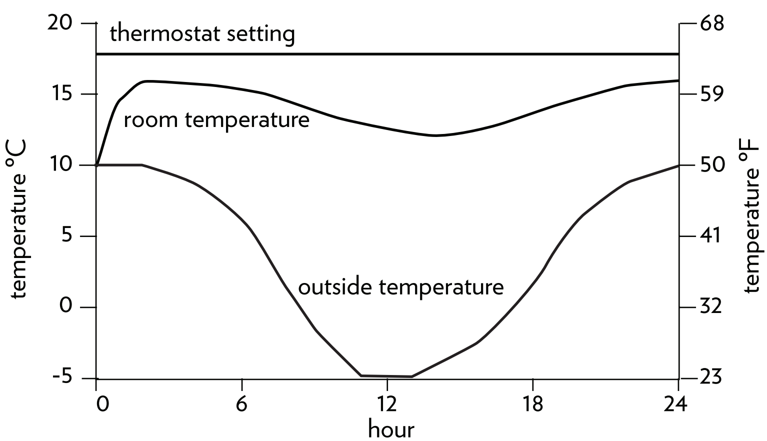


Figure 14. On a cold day, the furnace can't keep the room warm in this leaky house!

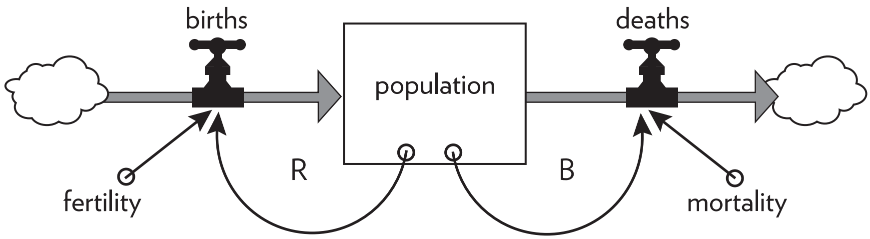
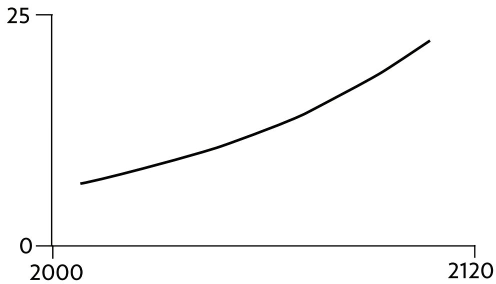
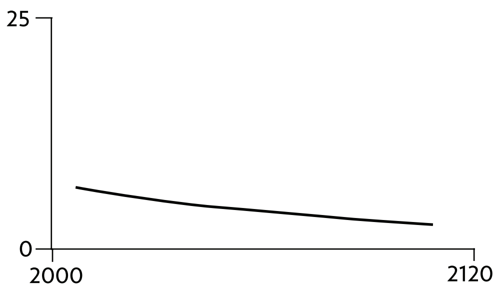


Figure 15. Population governed by a reinforcing loop of births and a balancing loop of deaths.

A: Growth



B: Decline



C: Stabilization

Figure 16. Three possible behaviors of a population: growth, decline, and stabilization.

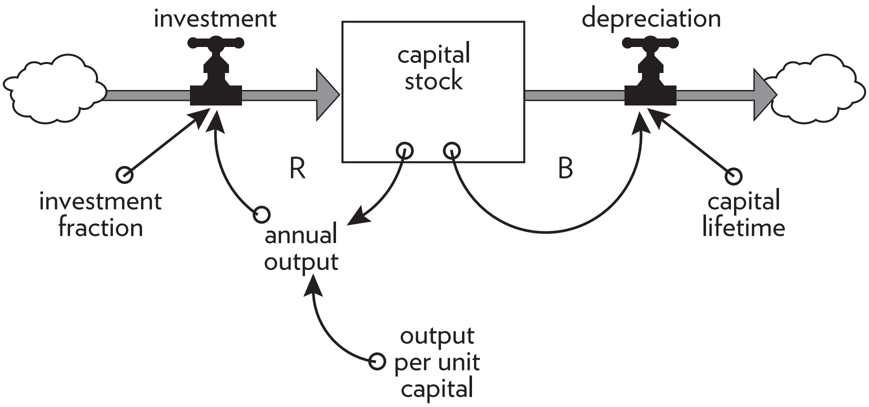


Figure 17. Like a living population, economic capital has a reinforcing loop (investment of output) governing growth and a balancing loop (depreciation) governing decline.

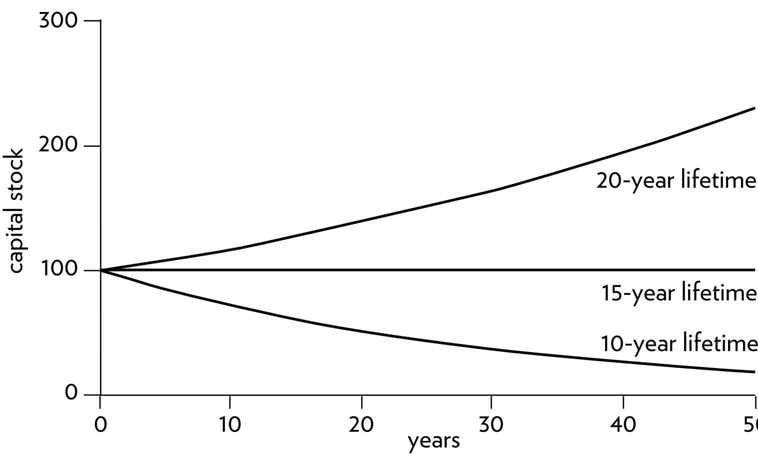


Figure 18. Growth in capital stock with changes in the lifetime of the capital. In a system with output per unit capital ratio of 1:3 and an investment fraction of 20 percent, capital with a lifetime of 15 years just keeps up with depreciation. A shorter lifetime leads to a declining stock of capital.

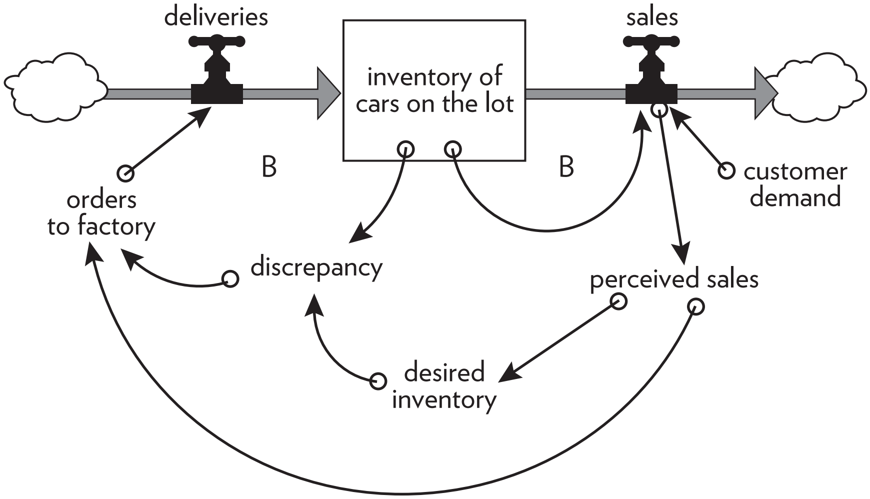


Figure 19. Inventory at a car dealership is kept steady by two competing balancing loops, one through sales and one through deliveries.

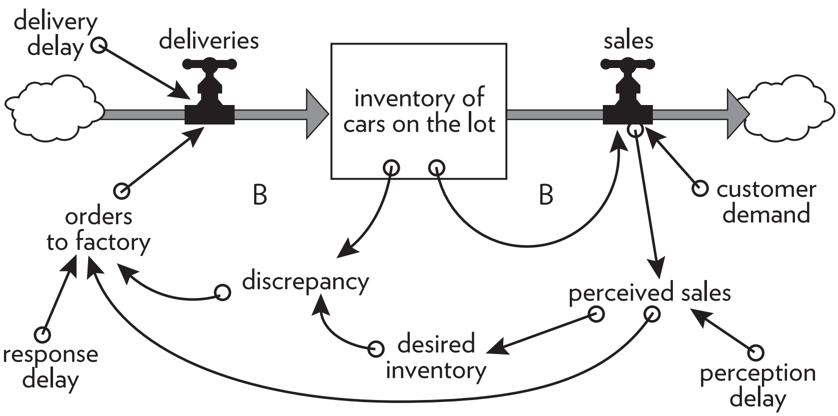
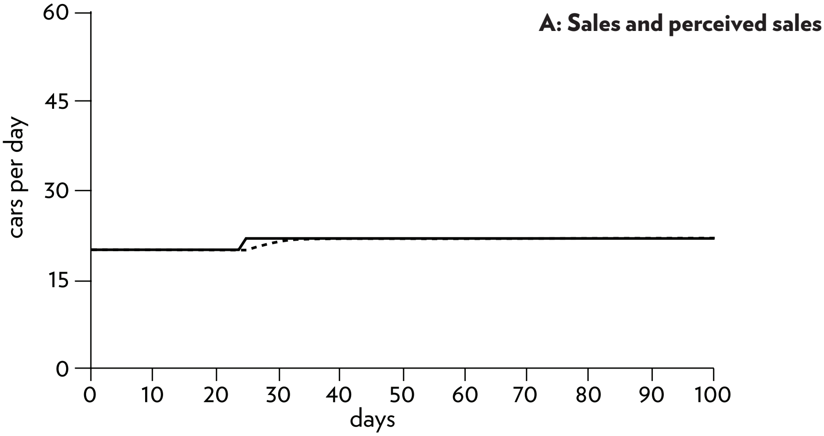


Figure 20. Inventory at a car dealership with three common delays now included in the picture—a perception delay, a response delay, and a delivery delay.



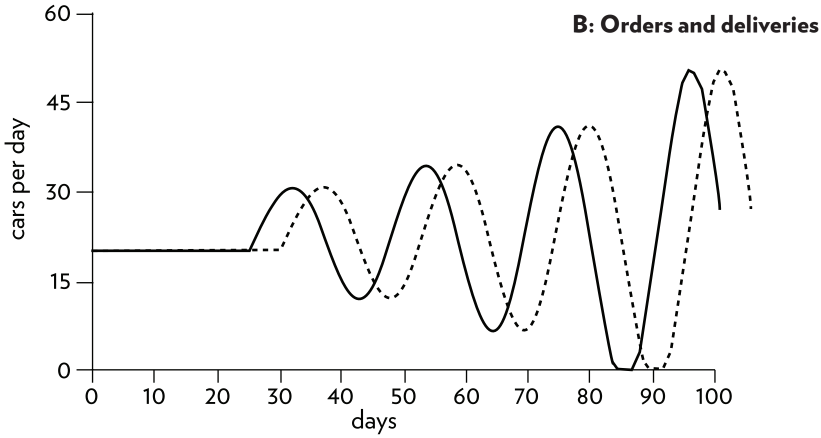


Figure 21. The response of orders and deliveries to an increase in demand. *A* shows the small but sharp step up in sales on day 25 and the car dealer's "perceived" sales, in which she averages the change over 3 days. *B* shows the resulting ordering pattern, tracked by the actual deliveries from the factory.

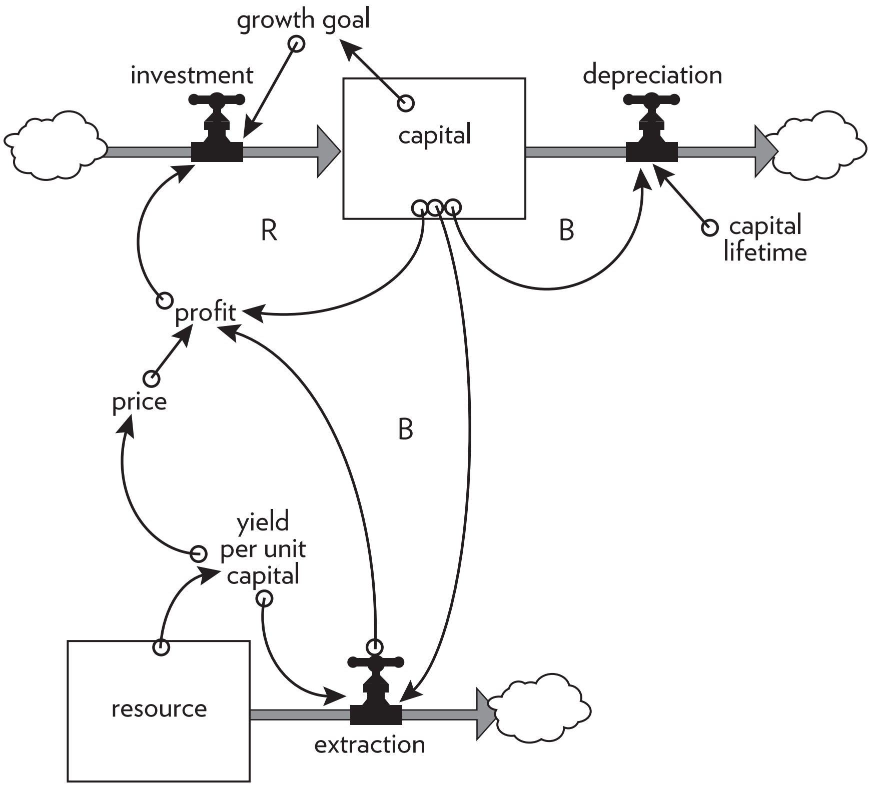


Figure 22. Economic capital, with its reinforcing growth loop constrained by a nonrenewable resource.

A: Extraction rate

B: Capital stock

C: Resource stock

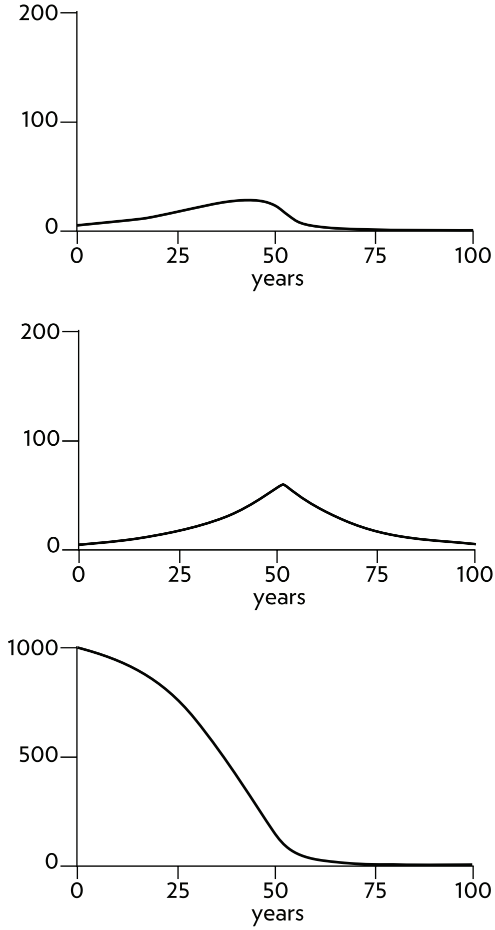


Figure 23. Extraction (A) creates profits that allow for growth of capital (B) while depleting the nonrenewable resource (C). The greater the accumulation of capital, the faster the resource is depleted.

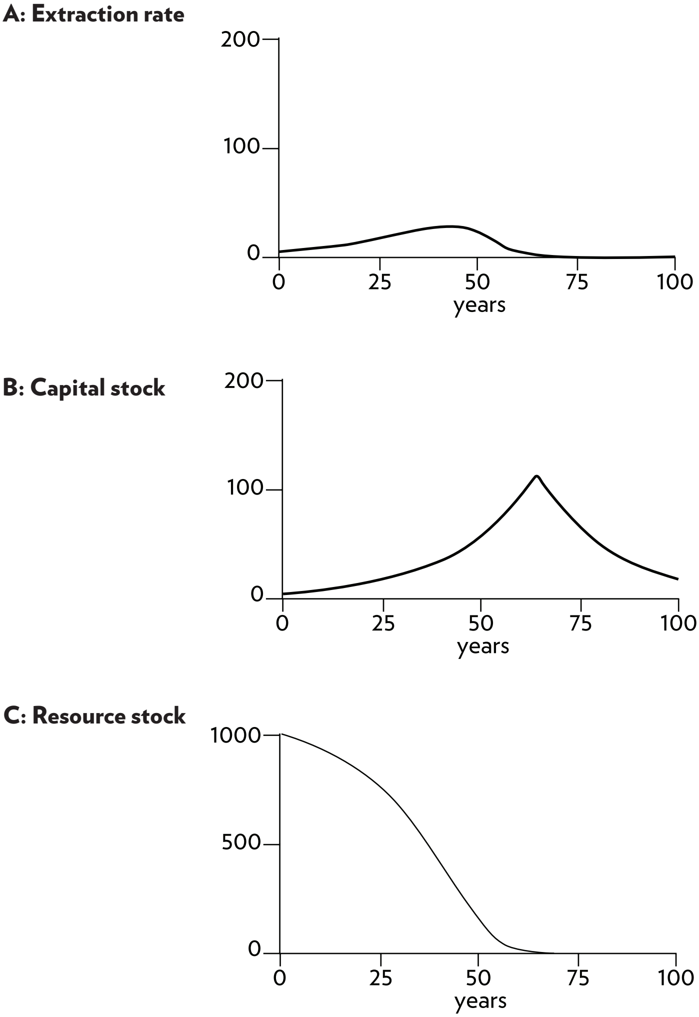


Figure 24. As price goes up with increasing scarcity, there is more profit to reinvest, and the capital stock can grow larger (B) driving extraction up for longer (A). The consequence is that the resource (C) is depleted even faster at the end.

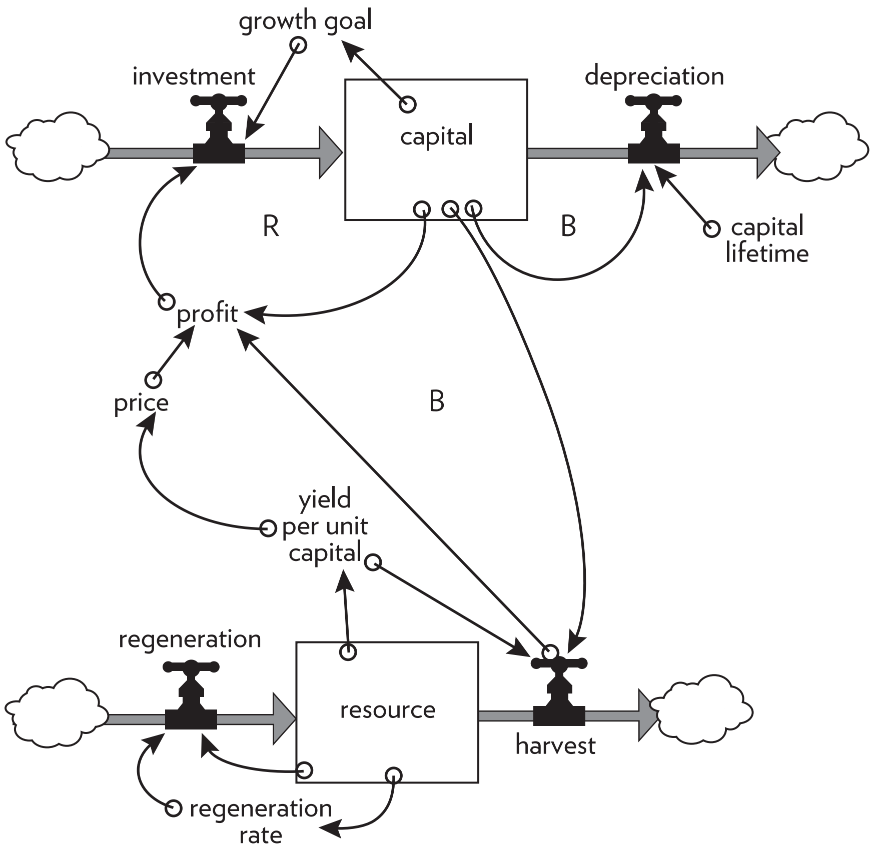


Figure 25. Economic capital with its reinforcing growth loop constrained by a renewable resource.

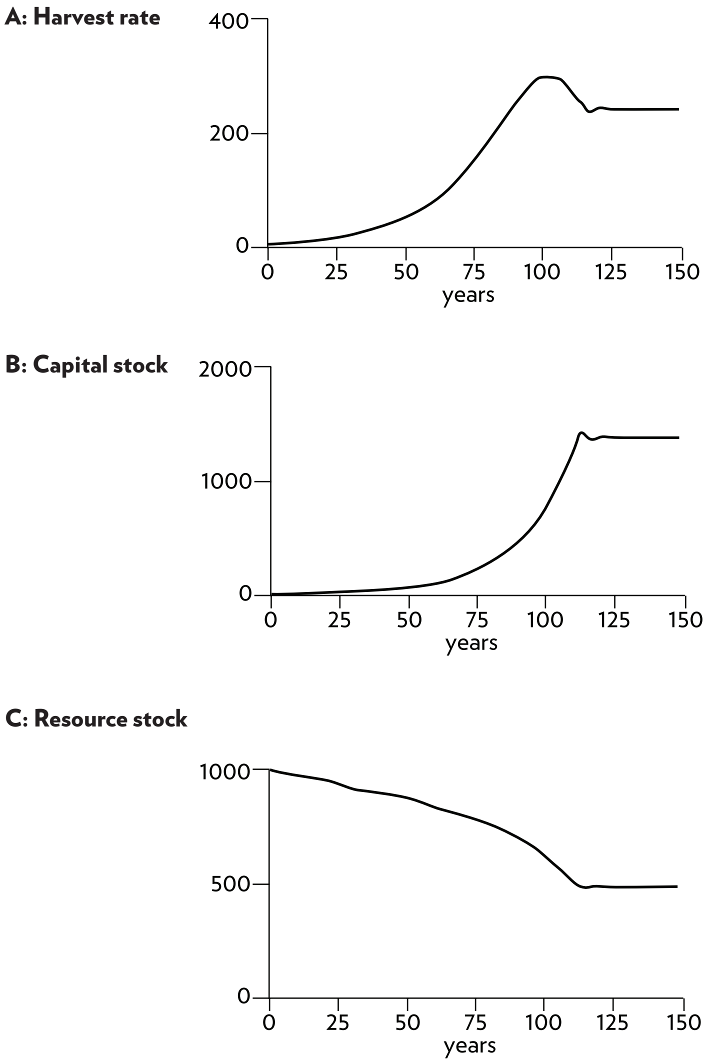


Figure 26. Annual harvest (A) creates profits that allow for growth of capital stock (B), but the harvest levels off, after a small overshoot in this case. The result of leveling harvest is that the resource stock (C) also stabilizes.

Appendix

System Definitions: A Glossary

Archetypes: Common system structures that produce characteristic patterns of behavior.

Balancing feedback loop: A stabilizing, goal-seeking, regulating feedback loop, also know as a "negative feedback loop" because it opposes, or reverses, whatever direction of change is imposed on the system.

Bounded rationality: The logic that leads to decisions or actions that make sense within one part of a system but are not reasonable within a broader context or when seen as a part of the wider system.

Dynamic equilibrium: The condition in which the state of a stock (its level or its size) is steady and unchanging, despite inflows and outflows. This is possible only when all inflows equal all outflows.

Dynamics: The behavior over time of a system or any of its components.

Feedback loop: The mechanism (rule or information flow or signal) that allows a change in a stock to affect a flow into or out of that same stock. A closed chain of causal connections from a stock, through a set of decisions and actions dependent on the level of the stock, and back again through a flow to change the stock.

Flow: Material or information that enters or leaves a stock over a period of time.

Hierarchy: Systems organized in such a way as to create a larger system. Subsystems within systems.

Limiting factor: A necessary system input that is the one limiting the activity of the system at a particular moment.

Linear relationship: A relationship between two elements in a system that has constant proportion between cause and effect and so can be drawn with a straight line on a graph. The effect is additive.

Nonlinear relationship: A relationship between two elements in a system where the cause does not produce a proportional (straight-line) effect.

Reinforcing feedback loop: An amplifying or enhancing feedback loop, also known as a "positive feedback loop" because it reinforces the direction of change. These are vicious cycles and virtuous circles.

Resilience: The ability of a system to recover from perturbation; the ability to restore or repair or bounce back after a change due to an outside force.

Self-organization: The ability of a system to structure itself, to create new structure, to learn, or diversify.

Shifting dominance: The change over time of the relative strengths of competing feedback loops.

Stock: An accumulation of material or information that has built up in a system over time.

Suboptimization: The behavior resulting from a subsystem's goals dominating at the expense of the total system's goals.

System: A set of elements or parts that is coherently organized and interconnected in a pattern or structure that produces a characteristic set of behaviors, often classified as its "function" or "purpose."

Summary of Systems Principles

Systems

- A system is more than the sum of its parts.
- Many of the interconnections in systems operate through the flow of information.
- The least obvious part of the system, its function or purpose, is often the most crucial determinant of the system's behavior.
- System structure is the source of system behavior. System behavior reveals itself as a series of events over time.

Stocks, Flows, and Dynamic Equilibrium

- A stock is the memory of the history of changing flows within the system.
- If the sum of inflows exceeds the sum of outflows, the stock level will rise.
- If the sum of outflows exceeds the sum of inflows, the stock level will fall.
- If the sum of outflows equals the sum of inflows, the stock level will not change — it will be held in dynamic equilibrium.

- A stock can be increased by decreasing its outflow rate as well as by increasing its inflow rate.
- Stocks act as delays or buffers or shock absorbers in systems.
- Stocks allow inflows and outflows to be de-coupled and independent.

Feedback Loops

- A feedback loop is a closed chain of causal connections from a stock, through a set of decisions or rules or physical laws or actions that are dependent on the level of the stock, and back again through a flow to change the stock.
- Balancing feedback loops are equilibrating or goal-seeking structures in systems and are both sources of stability and sources of resistance to change.
- Reinforcing feedback loops are self-enhancing, leading to exponential growth or to runaway collapses over time.
- The information delivered by a feedback loop—even nonphysical feedback—can affect only future behavior; it can't deliver a signal fast enough to correct behavior that drove the current feedback.
- A stock-maintaining balancing feedback loop must have its goal set appropriately to compensate for draining or inflowing processes that affect that stock. Otherwise, the feedback process will fall short of or exceed the target for the stock.
- Systems with similar feedback structures produce similar dynamic behaviors.

Shifting Dominance, Delays, and Oscillations

- Complex behaviors of systems often arise as the relative strengths of feedback loops shift, causing first one loop and then another to dominate behavior.
- A delay in a balancing feedback loop makes a system likely to oscillate.
- Changing the length of a delay may make a large change in the behavior of a system.

Scenarios and Testing Models

- System dynamics models explore possible futures and ask "what if" questions.
- Model utility depends not on whether its driving scenarios are realistic (since no one can know that for sure), but on whether it responds with a realistic pattern of behavior.

Constraints on Systems

- In physical, exponentially growing systems, there must be at least one reinforcing loop driving the growth *and* at least one balancing loop constraining the growth, because no system can grow forever in a finite environment.
- Nonrenewable resources are stock-limited.
- Renewable resources are flow-limited.

Resilience, Self-Organization, and Hierarchy

- There are always limits to resilience.
- Systems need to be managed not only for productivity or stability, they also need to be managed for resilience.
- Systems often have the property of self-organization—the ability to structure themselves, to create new structure, to learn, diversify, and complexify.
- Hierarchical systems evolve from the bottom up. The purpose of the upper layers of the hierarchy is to serve the purposes of the lower layers.

Source of System Surprises

- Many relationships in systems are nonlinear.
- There are no separate systems. The world is a continuum. Where to draw a boundary around a system depends on the purpose of the discussion.
- At any given time, the input that is most important to a system is the one that is most limiting.
- Any physical entity with multiple inputs and outputs is surrounded by layers of limits.
- There always will be limits to growth.

- A quantity growing exponentially toward a limit reaches that limit in a surprisingly short time.
- When there are long delays in feedback loops, some sort of foresight is essential.
- The bounded rationality of each actor in a system may not lead to decisions that further the welfare of the system as a whole.

Mindsets and Models

- Everything we think we know about the world is a model.
- Our models do have a strong congruence with the world.
- Our models fall far short of representing the real world fully.

Springing the System Traps

Policy Resistance

Trap: When various actors try to pull a system state toward various goals, the result can be policy resistance. Any new policy, especially if it's effective, just pulls the system state farther from the goals of other actors and produces additional resistance, with a result that no one likes, but that everyone expends considerable effort in maintaining.

The Way Out: Let go. Bring in all the actors and use the energy formerly expended on resistance to seek out mutually satisfactory ways for all goals to be realized—or redefinitions of larger and more important goals that everyone can pull toward together.

The Tragedy of the Commons

Trap: When there is a commonly shared resource, every user benefits directly from its use, but shares the costs of its abuse with everyone else. Therefore, there is very weak feedback from the condition of the resource to the decisions of the resource users. The consequence is overuse of the resource, eroding it until it becomes unavailable to anyone.

The Way Out: Educate and exhort the users, so they understand the consequences of abusing the resource. And also restore or strengthen the missing feedback link, either by privatizing the resource so each user feels the direct consequences of its abuse or (since many resources cannot be privatized) by regulating the access of all users to the resource.

Drift to Low Performance

Trap: Allowing performance standards to be influenced by past performance, especially if there is a negative bias in perceiving past performance, sets up a reinforcing feedback loop of eroding goals that sets a system drifting toward low performance.

The Way Out: Keep performance standards absolute. Even better, let standards be enhanced by the best actual performances instead of being discouraged by the worst. Set up a drift toward high performance!

Escalation

Trap: When the state of one stock is determined by trying to surpass the state of another stock—and vice versa—then there is a reinforcing feedback loop carrying the system into an arms race, a wealth race, a smear campaign, escalating loudness, escalating violence. The escalation is exponential and can lead to extremes surprisingly quickly. If nothing is done, the spiral will be stopped by someone's collapse—because exponential growth cannot go on forever.

The Way Out: The best way out of this trap is to avoid getting in it. If caught in an escalating system, one can refuse to compete (unilaterally disarm), thereby interrupting the reinforcing loop. Or one can negotiate a new system with balancing loops to control the escalation.

Success to the Successful

Trap: If the winners of a competition are systematically rewarded with the means to win again, a reinforcing feedback loop is created by which, if it is allowed to proceed uninhibited, the winners eventually take all, while the losers are eliminated.

The Way Out: Diversification, which allows those who are losing the competition to get out of that game and start another one; strict limitation on the fraction of the pie any one winner may win (antitrust laws); policies that level the playing field, removing some of the advantage of the strongest players or increasing the advantage of the weakest; policies that devise rewards for success that do not bias the next round of competition.

Shifting the Burden to the Intervenor

Trap: Shifting the burden, dependence, and addiction arise when a solution to a systemic problem reduces (or disguises) the symptoms, but does nothing to solve the underlying problem. Whether it is a substance that dulls one's perception or a policy that hides the underlying trouble, the drug of choice interferes with the actions that could solve the real problem.

If the intervention designed to correct the problem causes the self-maintaining capacity of the original system to atrophy or erode, then a destructive reinforcing feedback loop is set in motion. The system deteriorates; more and more of the solution is then required. The system will become more and more dependent on the intervention and less and less able to maintain its own desired state.

The Way Out: Again, the best way out of this trap is to avoid getting in. Beware of symptom-relieving or signal-denying policies or practices that don't really address the problem. Take the focus off short-term relief and put it on long-term restructuring.

If you are the intervenor, work in such a way as to restore or enhance the system's own ability to solve its problems, then remove yourself.

If you are the one with an unsupportable dependency, build your system's own capabilities back up before removing the intervention. Do it right away. The longer you wait, the harder the withdrawal process will be.

Rule Beating

Trap: Rules to govern a system can lead to rule-beating—perverse behavior that gives the appearance of obeying the rules or achieving the goals, but that actually distorts the system.

The Way Out: Design, or redesign, rules to release creativity not in the direction of beating the rules, but in the direction of achieving the purpose of the rules.

Seeking the Wrong Goal

Trap: System behavior is particularly sensitive to the goals of feedback loops. If the goals—the indicators of satisfaction of the rules—are defined inaccurately or incompletely, the system may obediently work to produce a result that is not really intended or wanted.

The Way Out: Specify indicators and goals that reflect the real welfare of the system. Be especially careful not to confuse effort with result or you will end up with a system that is producing effort, not result.

Places to Intervene in a System (in increasing order of effectiveness)

12. **Numbers:** Constants and parameters such as subsidies, taxes, and standards
11. **Buffers:** The sizes of stabilizing stocks relative to their flows
10. **Stock-and-Flow Structures:** Physical systems and their nodes of intersection
9. **Delays:** The lengths of time relative to the rates of system changes
8. **Balancing Feedback Loops:** The strength of the feedbacks relative to the impacts they are trying to correct
7. **Reinforcing Feedback Loops:** The strength of the gain of driving loops
6. **Information Flows:** The structure of who does and does not have access to information
5. **Rules:** Incentives, punishments, constraints
4. **Self-Organization:** The power to add, change, or evolve system structure
3. **Goals:** The purpose of the system
2. **Paradigms:** The mind-set out of which the system—its goals, structure, rules, delays, parameters—arises
1. **Transcending Paradigms**

Guidelines for Living in a World of Systems

1. Get the beat of the system.
2. Expose your mental models to the light of day.
3. Honor, respect, and distribute information.
4. Use language with care and enrich it with systems concepts.
5. Pay attention to what is important, not just what is quantifiable.
6. Make feedback policies for feedback systems.
7. Go for the good of the whole.

8. Listen to the wisdom of the system.
9. Locate responsibility within the system.
10. Stay humble—stay a learner.
11. Celebrate complexity.
12. Expand time horizons.
13. Defy the disciplines.
14. Expand the boundary of caring.
15. Don’t erode the goal of goodness.

About the Author

Donella Meadows (1941–2001) was a scientist trained in chemistry and biophysics (Ph.D., Harvard University). In 1970, she joined the Massachusetts Institute of Technology team lead by Dennis Meadows that produced "World3," a global computer model that explores the dynamics of human population and economic growth on a finite planet. In 1972, she was lead author of *The Limits to Growth,* the book that described for the general public the insights from the World3 modeling project. *Limits* was translated into twenty-eight languages and sparked debate around the world about the earth's carrying capacity and human choices. Meadows went on to write nine more books on global modeling and sustainable development and for fifteen years she wrote a weekly column, "The Global Citizen," reflecting on the state of our society and the complex connections in the world.

In 1991, Meadows was recognized as a Pew Scholar in Conservation and the Environment, and in 1994 she received a MacArthur Fellowship. She founded the Sustainability Institute in 1996 to apply systems thinking and organizational learning to economic, environmental, and social challenges. From 1972 until her death in 2001, Meadows taught in the Environmental Studies Program of Dartmouth College.

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